GIST: Weakly-Supervised Detection, Classification and Localization of GNSS Jamming and Spoofing from Crowdsourced ADS-B

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***Abstract*—GNSS interference has become a daily hazard for civil aviation, yet public monitors threshold a single integrity field per hexagon per day: they cannot tell jamming from spoofing, react with a delay of a day, and do not locate the emitter. We present GIST, which treats every aircraft as a remote GNSS sensor. GIST builds physics-consistency features that a spoofer controlling the GNSS solution cannot fake cheaply (geometric vs. barometric altitude relative to neighbours, readsb wind estimates vs. neighbour consensus, position jumps under nominal integrity, co-located aircraft at different altitudes), learns per-aircraft temporal representations with a Transformer, shares evidence between nearby aircraft with spatial graph attention, and localizes jammers with a differentiable Friis/radio-horizon likelihood combined with a learned heatmap. Labels come from a generative label model over physics-based labeling functions, so no manual annotation is required. On openly licensed ADS-B data (adsb.lol, ODbL) GIST reaches a spoofing recall of – at 1% false-positive rate on real 2026 traffic, where an integrity-threshold monitor reaches 0.01, and localizes simulated jammers among benign low-integrity aircraft with a median error of – km versus 42.5 km for non-linear least squares.**

I. Introduction

Global Navigation Satellite System (GNSS) interference affecting aviation has grown sharply since 2022 around the Baltic Sea, the Eastern Mediterranean, the Black Sea, the Middle East and the India–Pakistan border. Two very different threats are usually conflated: *jamming*, where the receiver loses the signal and the aircraft's broadcast Navigation Integrity Category (NIC) collapses, and *spoofing*, where a counterfeit signal yields a confident but false position. Because a spoofed receiver typically reports *high* integrity, monitors that threshold NIC [1] detect jamming well but are structurally blind to spoofing.

Every ADS-B equipped aircraft broadcasts its GNSS position, integrity indicators, barometric altitude, geometric (GNSS) altitude and velocity. Barometric altitude and air data are independent of GNSS, and a spoofer cannot make hundreds of aircraft mutually consistent. We exploit this redundancy.

Contributions. (1) A weakly-supervised, openly licensed dataset of jamming/spoofing labels built from physics-based labeling functions and a generative label model. (2) Spoofing-sensitive features that do not depend on integrity fields the spoofer controls. (3) GIST, a trajectory Transformer with spatial graph attention for 3-way classification. (4) A differentiable localization head trained sim-to-real on synthetic emitters injected into real traffic. (5) A minute-level detection latency, versus day-level maps.

II. Related Work

Localization from ADS-B. Liu et al. model jamming power with Friis propagation and solve for the emitter by non-linear least squares [2], [3], reaching ∼4 km on the Denver 2022 event. Schäfer et al. intersect radio horizons [4]. These methods need clean affected/unaffected labels and do not address spoofing. **Detection.** gpsjam.org [1] publishes a daily NIC-threshold hexagon map; space-based ADS-B has been used for global measurement [5], and industry white papers describe NIC/position cross-checks [6]. **Weak supervision.** We follow the data-programming paradigm [7] with a Dawid–Skene label model [8].

III. Data and Weak Supervision

A. Data

We use the adsb.lol `globe_history` archive [9] (ODbL-1.0): daily dumps of readsb [10] trace files containing every received position with source type (ADS-B, ADS-R, MLAT, TIS-B), barometric and geometric altitude, ground speed and track, and periodically attached detail objects with NIC, $\mathrm{NAC_p}$, SIL, GVA, true airspeed, heading and readsb's wind estimate. We use 5 days (1st of each month, January 2024 to October 2026), 67 M airborne trace points inside the study regions and 1.22 M aircraft-windows.

Points are binned into 30 s steps per aircraft and 10-minute windows of 20 steps. Six interference regions (BAL, EMED, BLK, MEA, INPK, MYN) and three comparison regions (Central Europe, US Midwest, SE Australia) are extracted.

B. Physics-consistency features

For every step we compute: integrity minima/maxima; kinematic residual $|\|\Delta\mathbf{p}\| - v_{gs}\,\Delta t|$ between consecutive

TABLE I
LABELING FUNCTIONS (WINDOW = 10 MIN). "MODERN" = ADS-B VERSION ≥ 1; "CRUISE" = MEDIAN ALTITUDE $> 10\,000$ FT.

LF	Vote	Condition
nic_zero	JAM	modern, cruise, $\geq 30\%$ steps NIC ≤ 1
nic_drop	JAM	modern, cruise, NIC ≥ 8 seen and $\geq 30\%$ steps NIC ≤ 6
pos_loss	JAM	cruise, ADS-B gap ≥ 90 s, $\geq 10\%$ steps NIC ≤ 6
cell_jam	JAM	$\geq 30\%$ of cell aircraft and $\geq 30\%$ own steps NIC ≤ 6
teleport	SPOOF	position jump > 10 km with NIC ≥ 7
gdiff	SPOOF	≥ 2 steps with $\lvert\Delta h_{geo-baro} - \text{nbr}\rvert > 1000$ ft, NIC > 7
cluster	SPOOF	member of a co-located, altitude-spread cluster
mlat†	SPOOF	MLAT vs. ADS-B position disagreement > 10 km
clean	NORMAL	all NIC ≥ 8, no jump/gdiff/cluster/MLAT flags
clean_lowcell	NORMAL	modern, cell NIC-low share $< 5\%$, own NIC never low

†held out of training labels; evaluation only.

ADS-B positions; ADS-B gaps and stale flags; geometric-minus-barometric altitude relative to the median of trusted aircraft (NIC ≥ 7) in the same H3 resolution-3 cell, 5-minute slot and 6000 ft band; deviation of readsb's wind estimate from the same consensus, using only points carrying a fresh detail object and excluding turning steps; and a spoof-cluster flag when ≥ 3 aircraft report positions within the same 0.005° cell within 60 s while their barometric altitudes span ≥ 5000 ft.

C. Labeling functions and label model

Ten labeling functions (LFs) vote NORMAL, JAMMED, SPOOFED or abstain per window (Table I). LFs never use the region name. A Dawid–Skene model estimates class-conditional LF confusion matrices by EM; windows on which every LF abstains are left unlabelled, since the all-abstain posterior otherwise favours the class with the highest abstention rate. The MLAT-consistency LF is held out of training and used only for evaluation.

IV. METHOD

A. Trajectory encoder

Each aircraft-window is a sequence of 20 step vectors (23 features incl. missingness flags), embedded by an MLP, given learned positional embeddings and a [CLS] token, and encoded by a 3-layer pre-norm Transformer ($d = 96$, 4 heads).

B. Spatial graph attention

Aircraft in a scene (±3° latitude, same 10-minute window, ≤ 256 aircraft) attend to each other within 400 km. Attention logits receive a learned bias from edge features $(\Delta x, \Delta y, d, \Delta h)$ in the spirit of GATv2 [11]; three layers.

C. Heads

A node head predicts the window class; a step head predicts the per-step class from the step token concatenated with the node embedding.

D. Differentiable localization

Given per-step jamming probabilities p_{it} at positions $\mathbf{x}_{it}$ and heights h_{it}, the physics layer scores each cell g of a 21×21 grid (40 km) and EIRP e by

$$\ell(g,e) = \sum_{i,t} p_{it} \log\sigma(z_{it}) + (1-p_{it}) \log\sigma(-z_{it}), \quad z_{it} = \frac{P_r(g,e,\mathbf{x}_{it},h_{it}) - T}{T} \tag{1}$$

with P_r from Friis free-space loss at L1, an elevation-dependent antenna gain and a 4/3-Earth radio horizon; $T = -115$ dBW [2] plus a learned offset and T learned. Logits are $\text{logsumexp}_e\, \ell(g,e)$ (normalised) plus a learned heatmap from grid-to-aircraft cross-attention; the location is the soft-argmax.

E. Training

Real scenes contribute soft node labels from the label model; simulated scenes (Sec. V) contribute exact node and step labels and the emitter position. Loss = weighted soft cross-entropy (node) + cross-entropy (step) + 0.5×(Gaussian-target cross-entropy + L1) (localization).

V. SIMULATION

Synthetic emitters are injected into clean real traffic at the step-feature level, then neighbour and cluster features are recomputed. Jammers (60%) have EIRP 20–50 dBW, ground (2–50 m) or airborne (1–10 km); receiver tolerance $\mathcal{N}(-115, 3^2)$ dBW; above tolerance NIC→0, geometric altitude is lost and positions drop with probability $\sigma(m/3)$; within 6 dB below, NIC$\in [1,6]$. Spoofers (40%, 10–40 dBW) capture receivers at margin > -5 dB (sticky) and replace positions with a static point, circle or drift, keep NIC≥ 8, offset geometric altitude and make reported velocity consistent with the fake track. Each training day contributes simulated scenes alongside real scenes.

VI. EXPERIMENTS

A. Setup

Days are split temporally: train up to 2025-06, validation 2025-07 to 2025-12, test 2026-01 to 2026-10. This gives 2 training, 1 validation and 2 test days.

B. Baselines

gpsjam-style cell threshold, Isolation Forest and LSTM autoencoder on the same windows; for localization, NIC-weighted centroid, radio-horizon scoring and Liu-style NLS (our re-implementation, 2.4 km median error on noise-free simulated ground jammers).

VII. CASE STUDIES

[TODO: BAL, EMED, INPK maps; KDEN once OpenSky access is granted]

TABLE II

TEST SPLIT (2026 DAYS). DETECTION METRICS PER AIRCRAFT-WINDOW: MACRO-F1 OVER 3 CLASSES, AVERAGE PRECISION (AP) AND SPOOFING RECALL AT 1% FALSE-POSITIVE RATE (R@1%). REAL = LABEL-MODEL TARGETS; SIM = EXACT SIMULATED TRUTH. SINGLE-SCORE BASELINES CANNOT SEPARATE JAMMING FROM SPOOFING; THEIR ANOMALY SCORE IS USED FOR BOTH. MEAN±STD OVER SEEDS.

	Real traffic				Simulated emitters			
Method	F1	Jam AP	Spoof AP	Spoof R@1%	F1	Jam AP	Spoof AP	Spoof R@1%
gpsjam-style threshold	0.62	0.86	0.01	0.01	0.26	0.93	0.07	0.01
Isolation Forest	0.56	0.80	0.04	0.06	0.27	0.48	0.15	0.01
LSTM autoencoder	0.38	0.37	0.03	0.06	0.27	0.58	0.13	0.03
GIST (0 seeds)	–	–	–	–	–	–	–	–

	Clean scenes		+5% benign low-NIC aircraft	
Jammer localization [km]	median	p95	median	p95
Centroid of affected	105.0	260.2	126.4	266.4
Radio-horizon score	121.3	382.3	138.0	448.6
NLS (Liu et al.)	26.0	301.7	42.5	315.4
GIST	–	–	–	–

TABLE III

ABLATIONS ON THE TEST SPLIT (SEED 0 UNLESS NOTED).

	Real		Sim		Loc.
Variant	F1	Spoof R@1%	F1	Spoof R@1%	med., km
GIST (full)	–	–	–	–	–
w/o spatial graph attention	–	–	–	–	–
integrity fields only	–	–	–	–	–
w/o simulated training scenes	–	–	–	–	–

VIII. LIMITATIONS AND ETHICS

Labels are weak: they encode our physics assumptions, and agreement with them is not ground truth; we therefore report results on simulated ground truth and on a held-out independent signal (MLAT). Simulation uses free-space propagation with a coarse antenna model. Coverage of crowdsourced receivers is uneven (sparse over parts of the Middle East and South Asia). All data are public broadcasts; we publish aggregates and events, not per-aircraft tracks. The work is defensive.

IX. CONCLUSION

[TODO: after results]

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